THE UNION OF FINITELY MANY NOWHERE DENSE SETS IS NOWHERE DENSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter, staged here as sources/stacks-topology/preamble.tex; this card uses retained-preamble.tex, which copies that shared preamble mechanically, including its reference/slogan/history display environments. Nothing was added, removed or redefined; the wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: the chapter carries no chapter-local bibliography and neither the counted statement nor its proof contains a citation, so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: the card was taken from the staged candidate list for this chapter, candidate lemma-nowhere-dense; its statement and proof spans are checked against the pinned chapter so that each span opens and closes a real, uncommented environment, and this lane confirms by rescanning that neither span carries a citation.

Allowed standard prerequisites (editorial).

- The definition of a nowhere dense subset as one whose closure has empty interior, and the behaviour of closures under finite unions. Set up in the author's section on nowhere dense sets, immediately before the counted statement. Fixes the notion to be preserved and the identity used in the proof.

Lemma 0.1. *Let X be a topological space. The union of a finite number of nowhere dense sets is a nowhere dense set.*

Proof. It suffices to prove the lemma for two nowhere dense sets as the result in general will follow by induction. Let $A, B \subset X$ be nowhere dense subsets. We have $\overline{A \cup B} = \overline{A} \cup \overline{B}$. Hence, if $U \subset \overline{A \cup B}$ is an open subset of X, then $U \setminus U \cap \overline{B}$ is an open subset of U and of X and contained in $\overline{A}$ and hence empty. Similarly $U \setminus U \cap \overline{A}$ is empty. Thus $U = \emptyset$ as desired. □

Date: stacks-project:topology.tex:local-snapshot-20261012; CC BY 4.0.